# Mayank Chandak

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## Education

**Carnegie Mellon University** **Aug. 2026 – May 2028**
*Master of Science in Robotic Systems Development* *Pittsburgh, PA*
*Relevant Coursework: Advanced Computer Vision, Robot Mobility, Robot Autonomy*

**Indian Institute of Technology Madras** **Jul. 2022 – May 2026**
*Bachelor of Technology in Mechanical Engineering, Minor in Artificial Intelligence and Machine Learning* *Chennai, India*
CGPA: **9.38/10** Top 5/205 Mechanical Engineering

## Experience

**Robotics Capstone Researcher, Nissan-Sponsored Project** **Aug. 2026 – Present**
*Vision-Based Learning for Humanoid Manipulation* *Carnegie Mellon University, Pittsburgh, PA*

- Architecting an end-to-end vision-to-action pipeline for a Unitree G1 humanoid, evaluating open-source VLAs including GR00T and $\pi$0.5 for fine-tuning on real-world Nissan manufacturing tasks.
- Designing a human-video-to-robot learning pipeline under limited demonstrations, exploring multi-view camera configurations, SAM 3D reconstruction, human-to-robot correspondence, and deployment-oriented policy learning.

**Undergraduate Researcher, INSPIRE Lab** **May 2025 – May 2026**
*Surgical Phase-Aware Reinforcement Learning* *IIT Madras, Chennai, India*

- Developed a phase-aware PPO framework for 4-DOF autonomous laparoscopic camera control, adapting centering, zoom, and motion objectives across surgical phases.
- Built a SOFA/LapGym simulation with deformable tissue, articulated instruments, FEM collision modeling, and pivotized endoscope control under RCM constraints; achieved **40% higher centering** and **67% lower motion jitter** than phase-blind RL.

**Young Research Fellow, Computational Imaging Lab** **Jul. 2024 – Mar. 2025**
*FlareGS: 4D Flare Removal for Urban Scenes* *IIT Madras, Chennai, India*

- Built the physics-based synthetic flare generation pipeline for FlareGS, contributing to a **1,000-pair** dataset across three controlled flare-severity regimes using real and synthetic urban driving scenes.
- Set up the end-to-end reconstruction pipeline, including **RGB/depth fusion, UFormer restoration, Gaussian initialization, camera-pose configuration, optimization losses, and 4D Gaussian Splatting training** for temporally consistent novel-view synthesis.
- Achieved **24.83/24.46 dB masked PSNR** at M2/M3 flare severity, improving over UFormer by **1.26/1.85 dB**; also contributed to VidFlareNet, reaching **20.03 PSNR** and **8.24 temporal-consistency loss**.

**ML Engineering Intern, Quantiphi** **Jun. 2024 – Jul. 2024**
*Generative AI and Document Intelligence* *Mumbai, India*

- Benchmarked Sparrow Parse and internal document-intelligence models for structured extraction from tabulated client documents, comparing model outputs across document-processing workflows.

## Publication

**M. Chandak***, S. S. T. Kuppa*, R. G. R. Matta, V. Gupta, K. Mitra, "FlareGS: 4D Flare Removal using Gaussian Splatting for Urban Scenes," *ICCV 2025 Workshop (2COOOL)*, 2025. *Equal contribution.

## Projects

**MyoFootball: Reward-Driven Musculoskeletal Control for Goal-Oriented Locomotion**

- Extended **MyoSuite's musculoskeletal MyoLeg environment** with ball dynamics and contact mechanics; designed a multi-stage reward hierarchy for locomotion, kicking, and balance.
- Iteratively tuned the reward formulation across randomized ball placements, achieving **80% goal-scoring accuracy** through extensive reward-design experiments.

**TD3 Reinforcement Learning for Robotic Arm Manipulation**

- Implemented TD3 from scratch in PyTorch for continuous-control door opening with a Franka Panda in Robosuite, including twin actor-critic networks, target networks, replay buffers, delayed updates, and target policy smoothing; systematically tuned learning rates, architectures, batch sizes, and update delays for training stability.

## Technical Skills

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| **Programming** | Python, C++, MATLAB, Bash/zsh |
| **Robotics & Simulation** | ROS 2, MuJoCo, SOFA, LapGym, Robosuite, MyoSuite, CoppeliaSim |
| **AI/ML** | PyTorch, TensorFlow, Stable-Baselines3, Reinforcement Learning, Deep Learning, Computer Vision |
| **Tools** | Git, GitHub, Linux/Ubuntu, OpenCV, NumPy, Pandas, SolidWorks, LaTeX |

## Honors & Achievements

- **Young Research Fellow Award**, IIT Madras: selected 1 of 17 students from 300+ applicants for funded undergraduate research.
- **Top-10 Global Rank**, PBDL sRGB Low-Light Enhancement Challenge.
- **Top 5% JEE Advanced** — **Top 1% JEE Main**.